

kshitij sinha

Durg, India

☎ 6268760622 ✉ kshitijsinha261@gmail.com 🌐 GitHub 🔗 LinkedIn 📄 Certificates

SKILLS

Databases :	MySQL, Firebase Firestore / Realtime Database
Frameworks Libraries :	PyTorch, NumPy, Pandas, Flask, Flutter
Programming Languages :	Python, Dart, SQL, HTML, R for data science, Data Structure and Algorithms, C, C++
Soft Skills :	Problem Solving, Team Collaboration, Analytical Thinking, Communication, Time Management
Tools Platforms :	Git, GitHub, Webots, Firebase, VS Code, Google Colab, Jupyter Notebook

PROJECTS

GPS-Denied Navigation System for Autonomous Drones | 🌐 Github

- Developed an autonomous drone navigation system for GPS-denied environments using Lazy Theta* path planning and real-time SLAM.
- Implemented 9-direction obstacle avoidance, autonomous replanning, and PID-based flight control to ensure safe and stable navigation in complex environments.
- Designed an interactive 3D dashboard for real-time path visualization, mission control, and drone telemetry monitoring.

Technologies / Tools Used : Python, Flask, Webots, SLAM, Lazy Theta*, PID Control, LiDAR, Three.js

Inventory Management System For University Services

- Designed and developed an inventory management system to improve selection, tracking, and documentation of stationery items across university facilities.
- Facilitated user ability to select multiple inventory items and automatically generate PDF reports.
- Constructed the application using Flutter (Dart), with Firebase and MySQL for backend support and data management.

Technologies / Tools Used : Firebase, MySQL, Flutter (Dart)

AI Based Sketch to UI Generator using Gemini | 🌐 Github

- Developed an AI-powered system to convert sketches/UI mockups into functional web interfaces.
- Integrated Google Gemini Vision for automated frontend code generation (HTML, CSS, JS).
- Implemented a multi-agent workflow for UI generation and code optimization.
- Enabled ZIP export and GitHub-ready deployment.

Technologies / Tools Used : Next.js, React, TypeScript, Tailwind CSS, FastAPI, Python, Google Gemini API, HTML, CSS, JavaScript

INTERNSHIP EXPERIENCE

Research Intern | Internship

Feb 2025 - Jul 2025

Indian Institute of Technology Bhilai (IIT Bhilai)

On-site • Bhilai, Chhattisgarh, India

- Implemented and evaluated 4 adversarial attacks, including FGSM, PGD, C&W, and Universal Perturbation, on deep learning models.
- Conducted targeted and untargeted adversarial attack analysis to evaluate model vulnerability and robustness in image-based tasks.
- Analyzed the robustness of 5 CNN architectures — VGG, ResNet, EfficientNet, DenseNet, and MobileNet — under adversarial conditions.
- Executed dataset preprocessing, model experimentation, and performance evaluation using Python and PyTorch during a 6-month research internship.

Technologies / Skills Used : python, pandas, NumPy, PyTorch, Deep Learning, Computer Vision, Adversarial Machine Learning, CNN Architectures

EDUCATION

UTD - Chhattisgarh Swami Vivekanand Technical University | Bhilai

Oct 2022 - Jun 2026

B.Tech (Hons.) in Computer Science and Engineering (Data Science)

CGPA : 7.42

Shakuntala Vidhyalaya , Bhilai (CBSE) | Bhilai

CGPA : 63.4%

Senior Secondary

Shakuntala Vidhyalaya , Bhilai (CBSE) | Bhilai

CGPA : 80%

Secondary

CERTIFICATIONS

Subject Matter Expert (Probability and Statistics)

Mar 2024 - Apr 2024

Chegg India Pvt. Ltd.

Management Information System

Jul 2025 - Oct 2025

NPTEL / SWAYAM

ACHIEVEMENTS

- Completed labs in prompt design using Vertex AI, developing Gen AI applications with Gemini and Streamlit during the Gen AI Study Jam at the University of Calcutta.
- Gained hands-on experience in machine learning research and problem-solving during a 6-month Summer Research Internship at IIIT Naya Raipur.
- Secured a Top 5 position in a 24-hour hackathon at NIT Raipur Techfest, advancing to the finals among multiple teams.
- Completed labs in prompt design using Vertex AI, developing Gen AI applications with Gemini and Streamlit during the Gen AI Study Jam at Rungta College of Engineering and Technology.
- Achieved 3rd position in an education and awareness program conducted by NSS.
- Participated in two 7-day unit camps organized by the National Service Scheme (NSS), demonstrating commitment to community service and teamwork.
- Contributed to the development of an AI-based Empathy and Moral Reflection System for criminal rehabilitation at HackJNU 4.0, Jawaharlal Nehru University, New Delhi.
- Achieved 2nd position in the National Science Day Competition organized by the University Teaching Department, CSVTU.
- Developed an AI-based project using Google Gemini at the GitHub Dev Days Hackathon at Rungta University.